Colormap/heatmap Chart

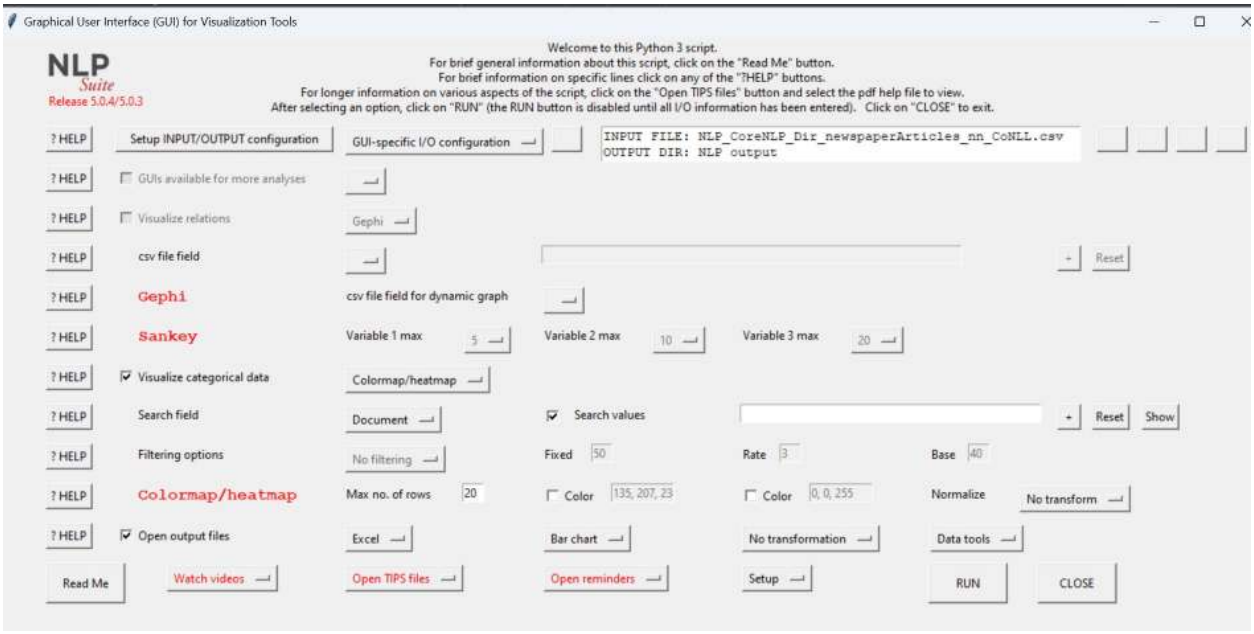
Contents

The Colormap/heatmap chart 1
The Colormap/heatmap chart GUI and input csv file 1

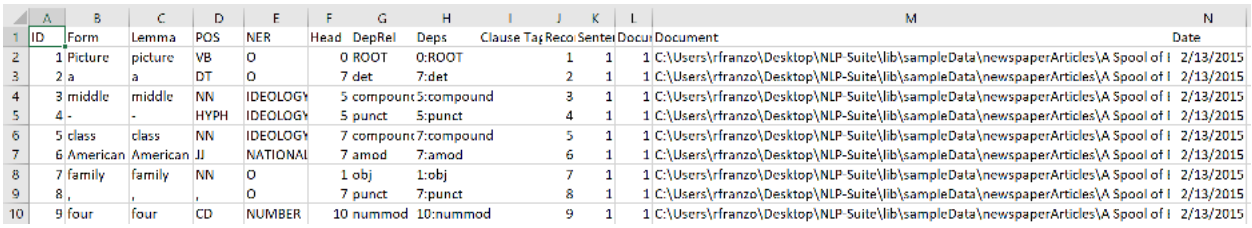
The Colormap/heatmap chart

A colormap, also called heatmap, is a two-dimensional data visualization technique that uses color to represent the magnitude of values in a dataset. The map represents two sets of values (X axis and Y axis) with colors. Typically using a single hue (e.g., blue), value frequencies change in color from lighter to darker hue. **Colormaps should be used for representing information that has ordering (e.g., frequencies).**

The Colormap/heatmap chart GUI and input csv file



Suppose that you have selected in input the CoNLL table csv file produced by the Stanford CoreNLP neural network parser with the newspaper articles stored in the sampleData subdirectory of the NLP Suite lib directory. The csv file has the following layout.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	ID	Form	Lemma	POS	NER	Head	DepRel	Deps	Clause Tag	Reco	Senter	Docu	Document	Date
2	1	Picture	picture	VB	O	0	ROOT	0:ROOT		1	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
3	2	a	a	DT	O	7	det	7:det		2	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
4	3	middle	middle	NN	IDEOLOGY	5	compound	5:compound		3	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
5	4	-	-	HYPH	IDEOLOGY	5	punct	5:punct		4	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
6	5	class	class	NN	IDEOLOGY	7	compound	7:compound		5	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
7	6	American	American	JJ	NATIONAL	7	amod	7:amod		6	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
8	7	family	family	NN	O	1	obj	1:obj		7	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
9	8	,	,	,	O	7	punct	7:punct		8	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015
10	9	four	four	CD	NUMBER	10	nummod	10:nummod		9	1	1	C:\Users\rfranzo\Desktop\NLP-Suite\lib\sampleData\newspaperArticles\A Spool of I	2/13/2015

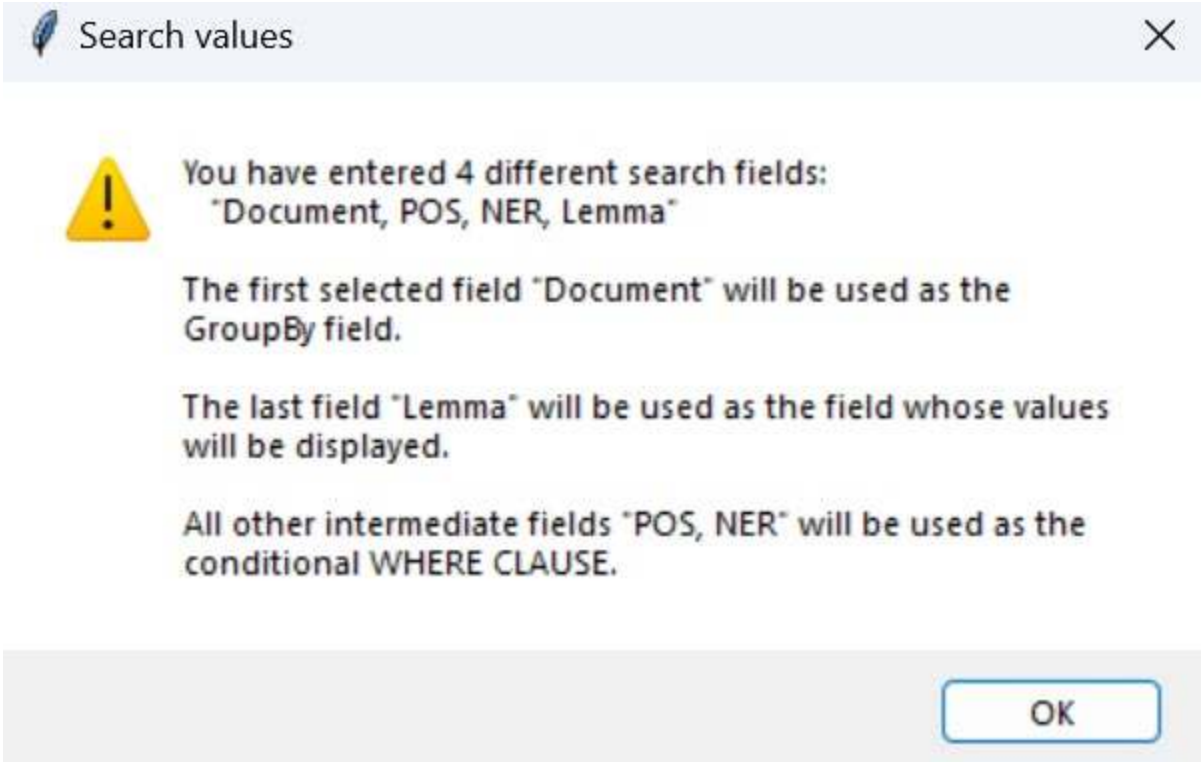
When you select the Colormap/heatmap visualization option, like all visualization options of the Suite, if the csv file contains a Document header, it will be automatically selected for convenience, since you may typically want to see how some field values change across documents.



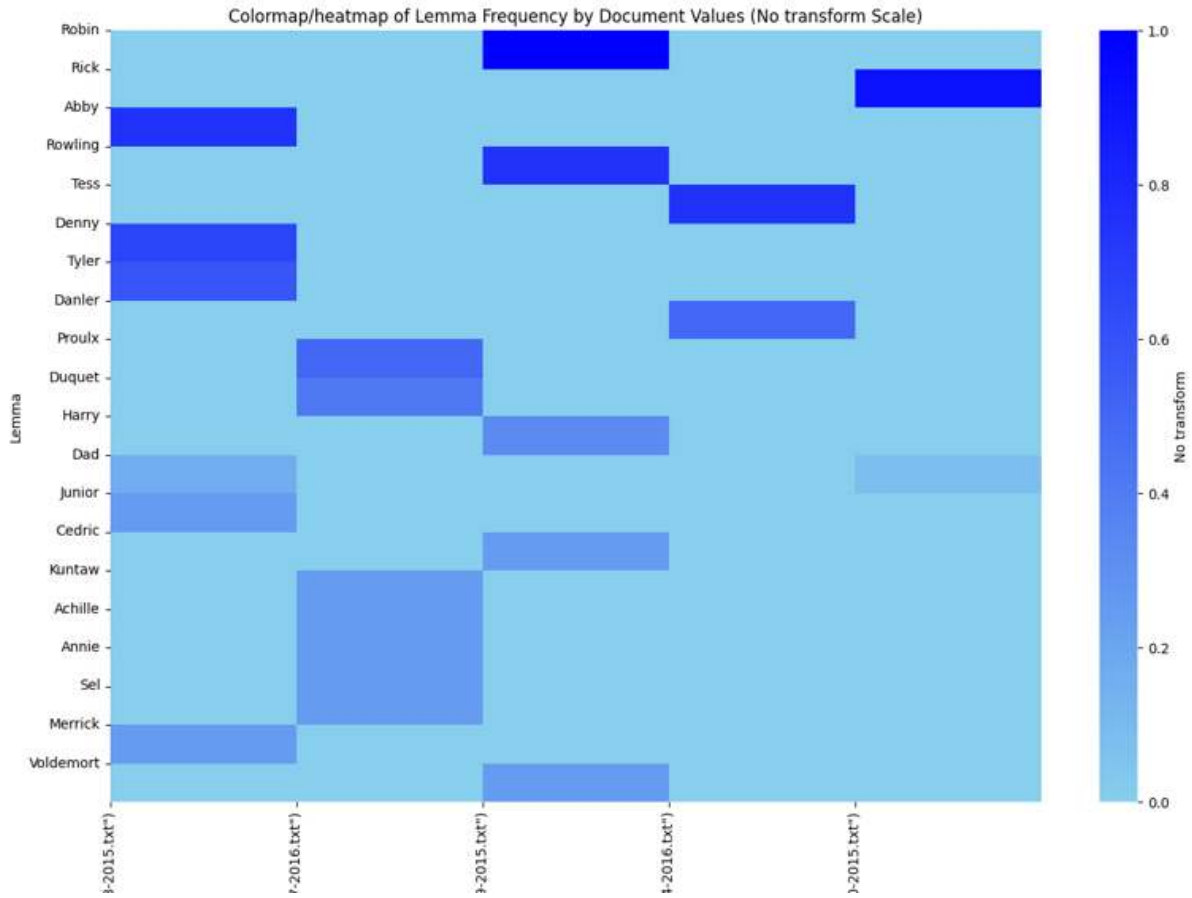
The colormap/heatmap option also allows you to set a limit to the number of rows you want to display not to clutter the chart with too many, perhaps, small values. **The rows displayed will contain the most frequent values.**

Suppose that you select the Document field, leaving blank the search values for all documents, then add the POS (Part of Speech) tag value of NNP (i.e., Proper noun), NER (Named Entity Recognition) tag of PERSON, and Lemma, leaving blank the search values for all Lemma values.

Upon clicking RUN, the algorithm will summarize your selection telling you how it will use your selection.

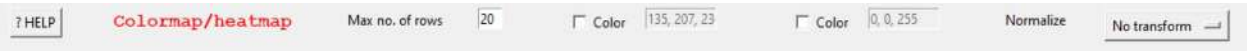


And here is the resulting chart, with default values (blue hue and top 20 rows only).



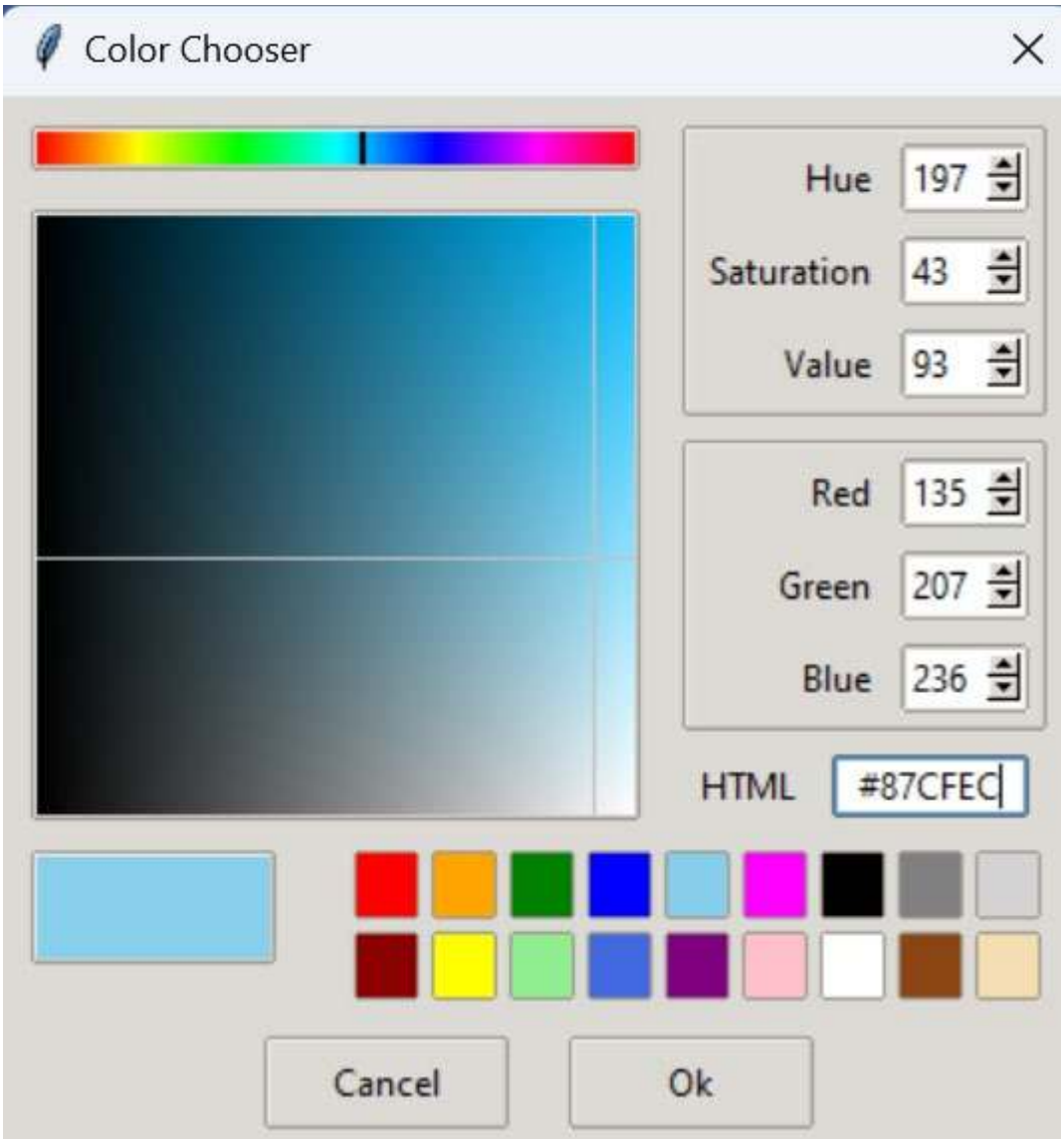
The darker blues tell you how frequently each Lemma word is used across the different documents.

The GUI also allows you to change the number of rows displayed, the color hue too be used, and whether any data transformation is requested.

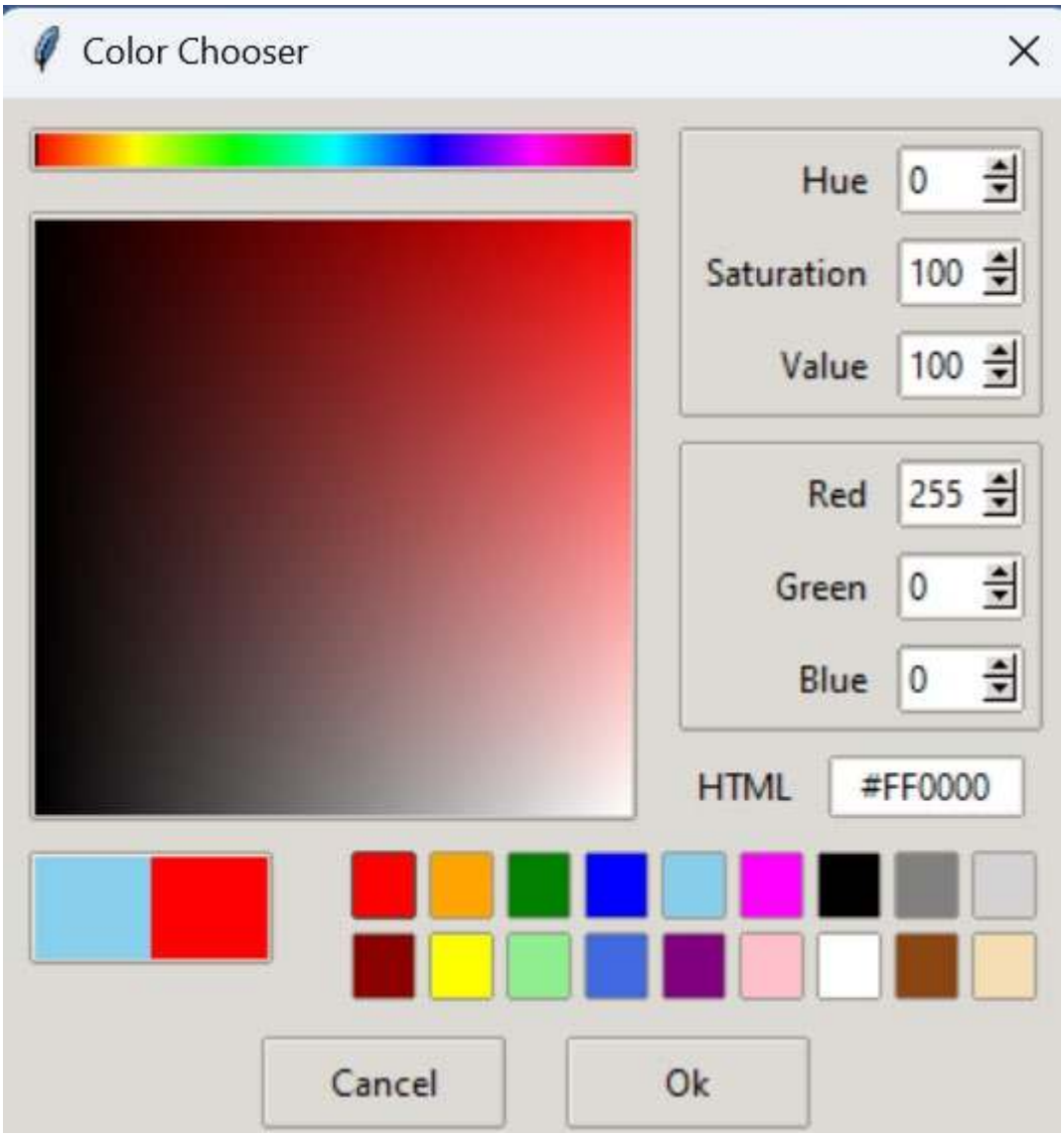


Of particular significance are the 2 Color checkboxes for lighter and darker colors to be used left to right on the chart.

Click on the lighter color checkbox. It will display the color picker GUI for you (blue as default).

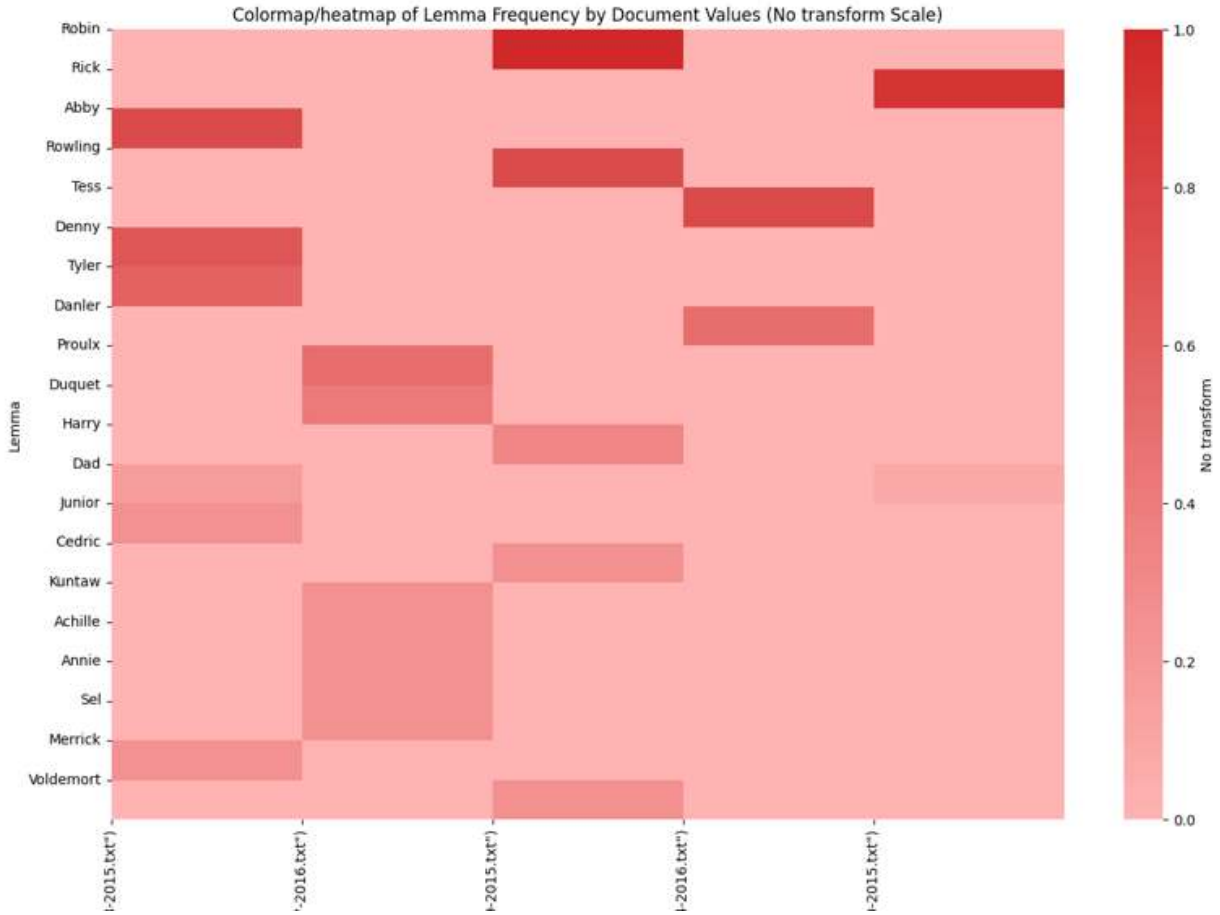


To change the default blue, click on any of the color buttons, e.g., red.



With our touchpad, select the part of the color area you want to use for the lighter hue, then click OK.

Do the same for the darker red hue.



In output you will also see a csv file with the dataframe values (colormap_dataframe.csv), where you can see the actual computed frequencies grouped by Document (or whichever group field you would have selected).

	A	B	C	D	E	F
1	Lemma	C:\Users\r	C:\Users\r	C:\Users\r	C:\Users\r	C:\Users\r
2	A.	0	0	0	1	0
3	Abby	9	0	0	0	0
4	Achille	0	3	0	0	0
5	Alfred	0	0	0	1	0
6	Amanda	1	0	0	0	0
7	Anne	2	0	0	0	0
8	Annie	0	3	0	0	0
9	Anthony	0	0	0	1	0